

Document Based Database

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R00225669

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MongoDB

Part 1 – MongoDB CRUD

1. Create a new database “banksdb” and new collection “clients” Manually insert all documents in the json file into this collection. List out all collections in your database and documents to confirm that you have all documents in the correct collection. Then insert the following client to your collection.

```
C:\Users\rlord>C:\Users\rlord\Downloads\mongosh-2.3.3-win32-x64\mongosh-2.3.3-win32-x64\bin\mongosh.exe
Current Mongosh Log ID: 682e633640a97103810d818f
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.3.3
Using MongoDB:      8.0.3
Using Mongosh:       2.3.3
mongosh 2.5.1 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2025-05-21T03:59:13.634+01:00: Access control is not enabled for the data
base. Read and write access to data and configuration is unrestricted
-----

test> use banksdb
switched to db banksdb
banksdb> show dbs
admin                40.00 KiB
config                72.00 KiB
local                80.00 KiB
rebe;_performance    108.00 KiB
rebel-fitness         20.00 KiB
rebel_performance    252.00 KiB
banksdb> db.createCollection("banks")
{ ok: 1 }
banksdb> show collections
banks
banksdb> db.banks.insertOne({"isActive": false, "balance": 3960.64, "age": 30,
"eyeColor": "blue", "name": "Dawn Keith", "gender": "female", "company": "COSMO
SIS", "email": "dawnkeith@cosmosis.com", "phone": "+1 (839) 437-3421", "address
": "392 Clifford Place, Fontanelle, Arizona, 2687"})
{
  acknowledged: true,
  insertedId: ObjectId('682e67e340a97103810d8190')
```

```
banksdb> banksdb> db.banks.insertMany([{"isActive": false, "balance": 1280.14, "age": 31, "eyeColor": "green", "name": "Bettie Eaton", "gender": "female", "company": "COMTREK", "email": "bettieeaton@comtrek.com", "phone": "+1 (861) 460-2317", "address": "203 Allen Avenue, Elrama, North Carolina, 4453"}, {"isActive": true, "balance": 2042.37, "age": 38, "eyeColor": "green", "name": "Margie Ayala", "gender": "female", "company": "VOIPA", "email": "margieayala@voipa.com", "phone": "+1 (941) 569-2231", "address": "111 Schroeders Avenue, Suitland, Louisiana, 7042"}, {"isActive": false, "balance": 3170.35, "age": 29, "eyeColor": "blue", "name": "John Doe", "gender": "male", "company": "ACME", "email": "john.doe@acme.com", "phone": "+1 (555) 123-4567", "address": "123 Main Street, New York, NY 10001"}])
banksdb> db.banks.countDocuments()
35
banksdb> db.banks.insertOne({"isActive": true, "balance": 2042.37, "age": 38, "eyeColor": "green", "name": "Margie Ayala", "gender": "female", "company": "VOIPA", "email": "margieayala@voipa.com", "phone": "+1 (941) 569-2231", "address": "111 Schroeders Avenue, Suitland, Louisiana, 7042"})
{
  acknowledged: true,
  insertedId: ObjectId('682e768540a97103810d81b3')
}
banksdb>
```

2. Explain why the profile of Margie has been inserted twice? How do we fix this problem? What happens if we try to reinsert the same profile of Margie again after you fixed the problem?

Margie Ayala was in the database twice because MongoDB doesn't stop duplicated documents by default. When Margie's information was entered twice with different ids, MongoDB didn't catch that it was the same person. To make sure any duplicates don't occur by making the email unique, MongoDB won't allow a person to be added if they share the same email.

If we try to add Margie's profile again after fixing the problem, MongoDB will show an error and it won't allow the duplicate to be added. This is good as it gets rid of having the Margie in the database more than once.

```
banksdb> db.banks.deleteOne({_id: ObjectId("682e768540a97103810d81b3")})
{ acknowledged: true, deletedCount: 1 }
banksdb> db.banks.createIndex({email: 1}, {unique: true})
email_1
banksdb> db.banks.findOne({name: "Margie Ayala"})
{
  _id: ObjectId('682e72a740a97103810d8192'),
  isActive: true,
  balance: 2042.37,
  age: 38,
  eyeColor: 'green',
  name: 'Margie Ayala',
  gender: 'female',
  company: 'VOIPA',
  email: 'margieayala@voipa.com',
  phone: '+1 (941) 569-2231',
  address: '111 Schroeders Avenue, Suitland, Louisiana, 7042'
}
banksdb>
```

3. Update the final balance after applying one year's 1% interest rate to the account of all active young female clients whose age below 30. [balance = principal * (1 + rate)]

```
banksdb> db.banks.find({gender:"female", age:{$lt:30}}, {name: 1, balance: 1
})
[
  {
    _id: ObjectId('682e72df40a97103810d8194'),
    balance: 1402.05,
    name: 'Annmarie Atkinson'
  },
  {
    _id: ObjectId('682e733a40a97103810d8197'),
    balance: 2284.89,
    name: 'Rachelle Chang'
  },
  {
    _id: ObjectId('682e736240a97103810d819e'),
    balance: 1579.74,
    name: 'Tommie Chase'
  },
  {
    _id: ObjectId('682e73cd40a97103810d81b0'),
    balance: 2041.17,
    name: 'Juana Puckett'
  },
  {
    _id: ObjectId('682e73e340a97103810d81b1'),
    balance: 3434.48,
    name: 'Juliette Slater'
  }
]
banksdb> |
```

```
banksdb> db.client.updateOne({name:"Annmarie Atkinson"}, {$set: {balance: 20
41.17 * 1.01}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
banksdb> db.client.updateOne({name:"Rachelle Chang"}, {$set: {balance: 2041.
17 * 1.01}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
banksdb> db.client.updateOne({name:"Tommie Chase"}, {$set: {balance: 2041.17
* 1.01}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
banksdb> db.client.updateOne({name:"Juana Puckett"}, {$set: {balance: 2041.1
7 * 1.01}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
banksdb> db.client.updateOne({name:"Juliette Slater"}, {$set: {balance: 2041
.17 * 1.01}})
{
  acknowledged: true,
```

4. All inactive middle aged male clients share some common hobbies “reading” and “video games” and they all love “Chinese”, “Korean”, and “French” food, keep this extra information in their profiles.

```
banksdb> db.banks.updateMany({ isActive: false, gender: "male", age: { $gte: 30, $lte: 60 } }, { $set: { hobbies: ["reading", "video games"], favoriteFoods: ["Chinese", "Korean", "French"]} })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 7,
  modifiedCount: 7,
  upsertedCount: 0
}
banksdb> |
```

```
banksdb> db.banks.find({ isActive: false, gender: "male", age: { $gte: 30, $lte: 60 } }, { name: 1, age: 1, hobbies: 1, favoriteFoods: 1, _id: 0 })
[
  {
    age: 37,
    name: 'Baker Townsend',
    favoriteFoods: [ 'Chinese', 'Korean', 'French' ],
    hobbies: [ 'reading', 'video games' ]
  },
  {
    age: 36,
    name: 'O'Donnell Rollins',
    favoriteFoods: [ 'Chinese', 'Korean', 'French' ],
    hobbies: [ 'reading', 'video games' ]
  },
  {
    age: 37,
    name: 'Howard Serrano',
    favoriteFoods: [ 'Chinese', 'Korean', 'French' ],
    hobbies: [ 'reading', 'video games' ]
  },
  {
    age: 32,
    name: 'Frost Mayer',
    favoriteFoods: [ 'Chinese', 'Korean', 'French' ],
    hobbies: [ 'reading', 'video games' ]
  },
  {
    age: 30,
    name: 'Garrison Richardson',
    favoriteFoods: [ 'Chinese', 'Korean', 'French' ],
    hobbies: [ 'reading', 'video games' ]
  },
  {
    age: 34,
    name: 'Burris Herman',
    favoriteFoods: [ 'Chinese', 'Korean', 'French' ],
    hobbies: [ 'reading', 'video games' ]
  },
  {
    age: 36,
    name: 'Osborne Guzman',
    favoriteFoods: [ 'Chinese', 'Korean', 'French' ],
    hobbies: [ 'reading', 'video games' ]
  }
]
```

5. Some of inactive male clients who have green eyes have lost their passion on Korean food, update their profiles to reflect this change.

```
banksdb> db.banks.updateMany({ isActive: false, gender: "male", eyeColor: "green" }, { $pull: { favoriteFoods: "Korean" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
banksdb> db.banks.find({ isActive: false, gender: "male", eyeColor: "green" }, { name: 1, favoriteFoods: 1, _id: 0 })
[
  { name: 'O'Donnell Rollins', favoriteFoods: [ 'Chinese', 'French' ] },
  { name: 'Howard Serrano', favoriteFoods: [ 'Chinese', 'French' ] },
  { name: 'Frost Mayer', favoriteFoods: [ 'Chinese', 'French' ] },
  {
    name: 'Garrison Richardson',
    favoriteFoods: [ 'Chinese', 'French' ]
  },
  { name: 'Osborne Guzman', favoriteFoods: [ 'Chinese', 'French' ] }
]
banksdb>
```

Part 2 – MongoDB Queries and Aggregation (json file)

1. List out all cities located in the southwest region of United States.

```
studentsdb> db.students.countDocuments()
133
studentsdb> db.students.distinct("city", {city: {$in: ["Austin", "San Diego", "Los Angeles", "Palm Springs", "Lahaina", "Sedona", "San Francisco", "Nashville", "Las Vegas", "Honolulu"]}})
[
  'Austin',
  'Honolulu',
  'Lahaina',
  'Las Vegas',
  'Los Angeles',
  'Nashville',
  'Palm Springs',
  'San Diego',
  'San Francisco',
  'Sedona'
]
studentsdb>
```

2. Find all adolescents aged under 20, lived in the Southern area of the United States, and have only one hobby. Evaluate the performance of this query, then add an index to improve the performance of this query.

```
studentsdb> db.students.find({age: { $lt: 20 }, city: { $in: ["San Antonio", "New Orleans", "St. Louis", "Branson", "Chicago", "Miami Beach", "Charleston", "Savannah", "Saint Augustine", "Orlando" ]}, hobbies: { $size: 1 }})

studentsdb> db.students.createIndex({ age: 1, city: 1, hobbies: 1 })
age_1_city_1_hobbies_1
```

3. How many female students passed all three assessment components in each region? The pass grade is 40 percent.

```
studentsdb> db.students.aggregate([{$match: {gender: "female", assessment1: { $gte: 40 }, assessment2: { $gte: 40 }, assessment3: { $gte: 40 }, }}, {$group: { _id: "$region", count: { $sum: 1 }}}])
```

4. How many students have passed all three assessment components in each class of each region?

```
studentsdb> db.students.aggregate([{$match: {assessment1: { $gte: 40 }, assessment2: { $gte: 40 }, assessment3: { $gte: 40 }, }}, {$group: { _id: { region: "$region", class: "$class" }, count: { $sum: 1 }}}])
```

5. What is the most popular hobby, and which class do students have it?

```
studentsdb> db.students.aggregate([{$unwind: "$hobbies" }, { $group: { _id: "$hobbies", count: { $sum: 1 }, classes: { $addToSet: "$class" } } }, { $sort: { count: -1 } }, { $limit: 1 }])
[ { _id: 'Watching Sports', count: 16, classes: [] } ]
```

Part 3 – MongoDB Queries and Aggregation (CSV file)

1. There are three groups of patients - patients who have cardiovascular symptom, patients who have neurological symptoms, patients who have sleep-related issues. Create a view for each group of patients. The main cardiovascular symptoms are chest pain, irregular heartbeat, swelling, excessive sweating, nausea/vomiting, and high-blood pressure. The main neurological symptoms are fatigue and weakness, dizziness, persistent cough, anxiety, pain in various parts of the body.

```
strokedb> db.createView("cardio_patients", "stroke_risk", [{ $match: { $or: [ { "Chest Pain": 1 }, { "Irregular Heartbeat": 1 }, { "Swelling (Edema)": 1 }, { "Excessive Sweating": 1 }, { "Nausea/Vomiting": 1 }, { "High Blood Pressure": 1 } ] } } ] })
{ ok: 1 }
```

```
strokedb> db.cardio_patients.find()
[
  {
    _id: ObjectId('682e9117b0b1442ca7f828a0'),
    Chest_Pain: 1,
    Shortness_of_Breath: 0,
    Irregular_Heartbeat: 0,
    'Fatigue_and_Weakness': 0,
    Dizziness: 1,
    'Swelling(Edema)': 1,
    'Pain_in_Neck/Jaw/Shoulder/Back': 0,
    Excessive_Sweating: 1,
    Persistent_Cough: 0,
    'Nausea/Vomiting': 1,
    High_Blood_Pressure: 0,
    'Chest_Discomfort (Activity)': 1,
    'Cold_Hands/Feet': 0,
    'Snoring/Sleep_Apnea': 1,
    'Anxiety/Feeling_of_Doom': 0,
    Age: 83,
    'Stroke_Risk(%)': 67.5,
    'At_Risk(Binary)': 1
  },
  {
    _id: ObjectId('682e9117b0b1442ca7f828a2'),
    Chest_Pain: 0,
    Shortness_of_Breath: 1,
    Irregular_Heartbeat: 1,
    'Fatigue_and_Weakness': 1,
    Dizziness: 1,
    'Swelling(Edema)': 1,
    'Pain_in_Neck/Jaw/Shoulder/Back': 0,
    Excessive_Sweating: 0,
    Persistent_Cough: 1,
    'Nausea/Vomiting': 1,
    High_Blood_Pressure: 0,
    'Chest_Discomfort (Activity)': 0,
    'Cold_Hands/Feet': 1,
    'Snoring/Sleep_Apnea': 0,
    'Anxiety/Feeling_of_Doom': 0,
    Age: 64,
    'Stroke_Risk(%)': 63,
    'At_Risk(Binary)': 1
  },
  {
    _id: ObjectId('682e9117b0b1442ca7f828a3'),
    Chest_Pain: 1,
    Shortness_of_Breath: 1,
    Irregular_Heartbeat: 1,
    'Fatigue_and_Weakness': 0,
    Dizziness: 1,
    'Swelling(Edema)': 0,
    'Pain_in_Neck/Jaw/Shoulder/Back': 0,
    Excessive_Sweating: 1,
    Persistent_Cough: 1,
    'Nausea/Vomiting': 1,
    High_Blood_Pressure: 1,
    'Chest_Discomfort (Activity)': 1,
    'Cold_Hands/Feet': 0,
    'Snoring/Sleep_Apnea': 0,
    'Anxiety/Feeling_of_Doom': 0,
    Age: 81,
    'Stroke_Risk(%)': 76.5,
    'At_Risk(Binary)': 1
  },
  {
    _id: ObjectId('682e9117b0b1442ca7f828a5'),
    Chest_Pain: 1,
    Shortness_of_Breath: 1,
    Irregular_Heartbeat: 1,
    'Fatigue_and_Weakness': 0,
    Dizziness: 0,
    'Swelling(Edema)': 1,
    'Pain_in_Neck/Jaw/Shoulder/Back': 0,
    Excessive_Sweating: 1,
    Persistent_Cough: 1,
    'Nausea/Vomiting': 1,
    High_Blood_Pressure: 0,
    'Chest_Discomfort (Activity)': 0,
    'Cold_Hands/Feet': 1,
    'Snoring/Sleep_Apnea': 0,
    'Anxiety/Feeling_of_Doom': 0,
    Age: 43,
    'Stroke_Risk(%)': 52.5,
    'At_Risk(Binary)': 1
  }
]
```

```
{
  _id: ObjectId('682e9117b0b1442ca7f828a6'),
  Chest_Pain: 1,
  Shortness_of_Breath: 1,
  Irregular_Heartbeat: 1,
  'Fatigue_and_Weakness': 1,
  Dizziness: 0,
  'Swelling(Edema)': 1,
  'Pain_in_Neck/Jaw/Shoulder/Back': 1,
  Excessive_Sweating: 0,
  Persistent_Cough: 1,
  'Nausea/Vomiting': 1,
  High_Blood_Pressure: 1,
  'Chest_Discomfort (Activity)': 1,
  'Cold_Hands/Feet': 1,
  'Snoring/Sleep_Apnea': 0,
  'Anxiety/Feeling_of_Doom': 0,
  Age: 40,
  'Stroke_Risk(%)': 66,
  'At_Risk(Binary)': 1
},
{
  _id: ObjectId('682e9117b0b1442ca7f828a7'),
  Chest_Pain: 1,
  Shortness_of_Breath: 1,
  Irregular_Heartbeat: 1,
  'Fatigue_and_Weakness': 1,
  Dizziness: 1,
  'Swelling(Edema)': 1,
  'Pain_in_Neck/Jaw/Shoulder/Back': 1,
  Excessive_Sweating: 1,
  Persistent_Cough: 1,
  'Nausea/Vomiting': 1,
  High_Blood_Pressure: 1,
  'Chest_Discomfort (Activity)': 0,
  'Cold_Hands/Feet': 0,
  'Snoring/Sleep_Apnea': 0,
  'Anxiety/Feeling_of_Doom': 1,
  Age: 87,
  'Stroke_Risk(%)': 94.5,
  'At_Risk(Binary)': 1
},
{
  _id: ObjectId('682e9117b0b1442ca7f828aa'),
  Chest_Pain: 1,
  Shortness_of_Breath: 0,
  Irregular_Heartbeat: 0,
  'Fatigue_and_Weakness': 0,
  Dizziness: 0,
  'Swelling(Edema)': 1,
  'Pain_in_Neck/Jaw/Shoulder/Back': 1,
  Excessive_Sweating: 1,
  Persistent_Cough: 0,
  'Nausea/Vomiting': 1,
  High_Blood_Pressure: 1,
  'Chest_Discomfort (Activity)': 1,
  'Cold_Hands/Feet': 1,
  'Snoring/Sleep_Apnea': 0,
  'Anxiety/Feeling_of_Doom': 0,
  Age: 63,
  'Stroke_Risk(%)': 62.5,
  'At_Risk(Binary)': 1
},
{
  _id: ObjectId('682e9117b0b1442ca7f828af'),
  Chest_Pain: 1,
  Shortness_of_Breath: 0,
  Irregular_Heartbeat: 0,
  'Fatigue_and_Weakness': 1,
  Dizziness: 0,
  'Swelling(Edema)': 1,
  'Pain_in_Neck/Jaw/Shoulder/Back': 1,
  Excessive_Sweating: 0,
  Persistent_Cough: 1,
  'Nausea/Vomiting': 1,
  High_Blood_Pressure: 0,
  'Chest_Discomfort (Activity)': 1,
  'Cold_Hands/Feet': 1,
  'Snoring/Sleep_Apnea': 1,
  'Anxiety/Feeling_of_Doom': 0,
  Age: 66,
  'Stroke_Risk(%)': 69,
  'At_Risk(Binary)': 1
},
}
```

```
strokedb> db.createView("neuro_patients", "stroke_risk", [{ $match: { $or: [{ "Fatigue & Weakness": 1 }, { "Dizziness": 1 }, { "Persistent Cough": 1 }, { "Anxiety/Feeling of Doom": 1 }, { "Pain in Neck/Jaw/Shoulder/Back": 1 } ] } } ] }
{ ok: 1 }
strokedb> db.neuro_patients.find().limit(2)
[
  {
    _id: ObjectId('682e9117b0b1442ca7f828a0'),
    Chest_Pain: 1,
    Shortness_of_Breath: 0,
    Irregular_Heartbeat: 0,
    'Fatigue_and_Weakness': 0,
    Dizziness: 1,
    'Swelling(Edema)': 1,
    'Pain_in_Neck/Jaw/Shoulder/Back': 0,
    Excessive_Sweating: 1,
    Persistent_Cough: 0,
    'Nausea/Vomiting': 1,
    High_Blood_Pressure: 0,
    'Chest_Discomfort (Activity)': 1,
    'Cold_Hands/Feet': 0,
    'Snoring/Sleep_Apnea': 1,
    'Anxiety/Feeling_of_Doom': 0,
    Age: 83,
    'Stroke_Risk(%)': 67.5,
    'At_Risk(Binary)': 1
  },
  {
    _id: ObjectId('682e9117b0b1442ca7f828a2'),
    Chest_Pain: 0,
    Shortness_of_Breath: 1,
    Irregular_Heartbeat: 1,
    'Fatigue_and_Weakness': 1,
    Dizziness: 1,
    'Swelling(Edema)': 1,
    'Pain_in_Neck/Jaw/Shoulder/Back': 0,
    Excessive_Sweating: 0,
    Persistent_Cough: 1,
    'Nausea/Vomiting': 1,
    High_Blood_Pressure: 0,
    'Chest_Discomfort (Activity)': 0,
    'Cold_Hands/Feet': 1,
    'Snoring/Sleep_Apnea': 0,
    'Anxiety/Feeling_of_Doom': 0,
    Age: 64,
    'Stroke_Risk(%)': 63,
    'At_Risk(Binary)': 1
  }
]
```

```
strokedb> db.createView("sleep_patients", "stroke_risk", [{ $match: { "Snoring/Sleep Apnea": 1 } } ] }
{ ok: 1 }
strokedb> db.sleep_patients.find().limit(2)
```

2. Dr. James Smith is a cardiologist, he wants to have an account to directly access the information of patients who suffer from cardiovascular symptoms. He wants to know how many young patients are under 30 years old, who have suffered from chest pain, irregular heartbeats, high blood pressure, and have over 40% chance of getting a stroke.

```
strokedb> db.stroke_risk.countDocuments({Age:{$lt:30},"Chest Pain":1,"Irregular Heartbeat":1,"High Blood Pressure":1,"Stroke Risk (%)":{$gt:40}})
0
strokedb> |
```

3. Switch back to the normal user account, how many pensioners aged between 60 and 70 do not have a risk to get a stroke even though they constantly feel dizzy, fatigue, and aches everywhere.

```
strokedb> db.stroke_risk.countDocuments({Age:{$gte:60,$lte:70},"At Risk (Binary)":0,"Dizziness":1,"Fatigue & Weakness":1,"Pain in Neck/Jaw/Shoulder/Back":1})
0
```

4. Stay as a normal user account, how many patients aged between 40 and 50 have over 45% chance of not getting a stroke when they snore heavily.

```
strokedb> db.stroke_risk.countDocuments({Age:{$gte:40,$lte:50},"Stroke Risk (%)":{$lt:55},"Snoring/Sleep Apnea":1})
0
```

5. Stay as a normal user account, find the chances of getting a stroke and the age of a patient when s/he participates some activities without any anxieties, s/he constant feels discomfort in his/her chest, hands/feet are cold, and sometimes short of breath.

```
strokedb> db.stroke_risk.find({"Chest Discomfort (Activity)":1,"Cold Hands/Feet":1,"Shortness of Breath":1,"Anxiety/Feeling of Doom":0}, {Age:1,"Stroke Risk (%)":1,_id:0})
```